LLL-C-251A
August 26, 1969
SUPERSEDING
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1 August 1933

FEDERAL SPECIFICATION

CHARCOAL, WOOD

This specification was approve by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers charcoal of one grade only.

1.2 Classification. Charcoal shall be of the following types as specified (see 6.2):

Type I - Lump.
Type II - Briquet.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-S-50 - Sacks, Shipping Paper, Reinforced.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.

FSC 9110

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specification and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40 inch x 48 inch Pallets. Skids, Runners, or Pallet-Type Base.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Classification Board:

National Motor Freight Classification.

(Application for copies should be addressed to the National Classification Board, 1616 P Street, N. W., Washington, D. C. 20036.)

Uniform Classification Committee:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. Charcoal consist of the product made by charring hard or soft wood.

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3.2 Physical characteristics.

3.2.1 Contamination. Incompletely charred wood, trash, and other material foreign to charcoal produced shall not exceed 1 percent by weight.

3.2.2 Moisture. The moisture content (by weight) shall not be in excess of 5 percent.

3.2.3 Volatile matter. Volatile matter content shall not be in excess of 20 percent by weight.

3.2.4 Type I. Type I, lump form charcoal, shall be black, nonlustrous, with glossy conchoidal fracture.

3.2.4.1 Size. A minimum of 90 percent, by weight, of the type I charcoal shall remain on a 2-1/2 inch screen, and a maximum of 2 percent, by weight, shall pass through a 1-1/4 inch screen.

3.2.4.2 Ash. The ash content of type I charcoal shall not exceed 4 percent when tested as specified in 4.4.2.3.3.

3.2.5 Type II. Type II, briquet form charcoal, shall be pulverized or granulated charcoal which is compressed and held together by binder.

3.2.5.1 Size. A minimum of 90 percent, by weight, of the type II charcoal shall remain on a 1-1/4 inch screen, and a maximum of 2 percent by weight, shall pass through a 3/4 inch screen.

3.2.5.2 Ash. The ash content of type II charcoal shall not exceed 6 percent when tested as specified in 4.4.2.3.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

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4.3 Quality conformance inspection.

4.3.1 Lot. Unless otherwise specified, each shipment shall be divided into lots of 200 sacks.

4.3.2 Sampling. One sack shall be selected at random from each lot for examination and test.

4.3.3 Examination. Samples selected in accordance with 4.3.2 shall be examined in accordance with 4.4.1. Presence of one or more defects shall be cause for rejection.

4.3.4 Tests. Samples selected in accordance with 4.3.2 shall be tested as specified in 4.4.2. Failure of any test shall be cause for rejection of the lot.

4.4 Inspection procedure.

4.4.1 Examination. The charcoal shall be examined for the following defects:

101. Color not as specified.
102. Type I charcoal not as specified
103. Type II charcoal not as specified.
104. Incompletely charred wood, trash, and other foreign matter in excess of 1 percent by weight.

4.4.2 Tests.

4.4.2.1 Preparation of sample. The sack shall be dumped and quartered. Two opposite quarters shall be collected for the screen test. The charcoal remaining shall be again quartered. Two opposite quarters shall be collected for the other tests.

4.4.2.2 Screen analysis.

4.4.2.2.1 Type I. The sample of type I charcoal shall be accurately weighed. The sample shall then be sieved through a 2-1/2 inch screen in such increments as will allow the pieces to be in direct contact with the meshes after completion of the shaking of each increment. Each piece of charcoal retained on the screen shall be upended on the screen to determine if it will in any position pass through the screen. Failure to meet the minimum requirement of 90 percent (by weight) retention on the 2-1/2 inch screen shall constitute failure of the test. The same process as outlined above shall be conducted using a 1-1/4 inch screen. Passage of more than 2 percent (by weight) of the sample through the screen, shall constitute failure of the test.

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4.4.2.2.2 Type II. The sample of type II charcoal shall be tested for size as outlined in 4.4.2.2.1 except that 1-1/4 inch and 3/4 inch screens respectively, shall be used. Nonconformance to 3.2.5.1 shall constitute failure of this test.

4.4.2.3 Moisture, volatile matter, and ash determination. The sample prepared in accordance with 4.4.2.1 shall be rapidly pulverized in a mill until the entire sample will pass a No. 20 sieve. Long grinding time shall be avoided. The sample shall be thoroughly mixed and 50 grams shall be stored in an airtight container. The remainder of the sample shall be discarded.

4.4.2.3.1 Moisture. A muffle furnace shall be heated to 750°C. plus or minus 5°C. and two previously ignited porcelain crucibles with covers shall be placed in the furnace for a minimum of 10 minutes. The crucibles shall then be be cooled in a desiccator, containing calcium chloride as a drying agent, for not less than 1 hour. The weight of each crucible shall be recorded to the nearest 0.1 milligram. One gram plus or minus 0.1 milligram of sample prepared in accordance with 4.4.2.3 shall be added to each crucible. The samples shall be placed in an oven at 105°C. plus or minus 2°C. for 2 hours plus or minus 10 minutes. The crucibles with dried samples shall be removed from the oven and placed in a desiccator for a minimum of 1 hour. The weight of each sample shall be recorded to the nearest 0.1 milligram. The procedure of exposure to 105°C. cooling in a desiccator and weighing shall be repeated until the change in weigh does not exceed 0.0005 gram. The percentage of moisture in each sample shall be calculated as follows:

$$\text{Moisture, percent} = \frac{A - B}{A} \times 100.$$

Where:

A = Weight of sample used in grams.
B = weight of sample after drying at 105°C., in grams.

Nonconformance to 3.2.2 shall constitute failure of this test.

4.4.2.3.2 Volatile matter. The muffle furnace shall be heated to 950°C. plus or minus 5°C. The crucibles used for the moisture determination, with lids in place and containing the samples, shall be heated as follows: Two minutes on the outer ledge of the furnace (approximately 300°C.) and then approximately 3 minutes on the edge of the furnace (approximately 500°C.) with the door open. The samples shall then be moved to the rear of the furnace, the door closed, and the samples shall be heated to 950°C. plus or minus 5°C. for 6 minutes plus or minus 10 seconds. The samples shall be watched through a small peep-hole in the muffle door. If sparking occurs, results will be in error and new samples shall be prepared and dried in accordance with 4.4.2.3 and 4.4.2.3.1. After exposure to 950°C., the sample shall be cooled in a desiccator for 1 hour and weighed. The percentage of volatile matter shall be calculated as follows:

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$$\text{Volatile matter, percent} = \frac{B - C}{B} \times 100$$

Where:

B = weight of sample after drying at 105°C. in accordance with 4.4.2.3.1.
C = weight of sample after heating at 950°C. for 6 minutes, in grams.

Nonconformance to 3.2.3 shall constitute failure of this test.

4.4.2.3.3 Ash. The lids and the uncovered crucibles containing the samples used in the volatile matter determination (see 4.4.2.3.2) shall be placed in a muffle furnace at 750°C. plus or minus 5°C. for a minimum of 6 hours. The crucibles shall be cooled in a desiccator for 1 hour and weighed to the nearest 0.1 milligram. The burning, cooling, and weighing of the samples shall be repeated until a succeeding cycle results in a loss of weight that is less than 0.0005 gram. Calculate the percentage of ash in the sample as follows:

$$\text{Ash, percent} = \frac{D}{B} \times 100$$

Where:

D = grams of residue after burning.
B = weight of sample after drying at 105°C. in accordance with 4.4.2.3.1.

Nonconformance to 3.2.4.2 or 3.2.5.2, as applicable, shall constitute failure of this test.

4.5 Inspection of preparation for delivery.

4.5.1 Quality conformance inspection pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completely processed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

105. Materials and containers not as specified. Each incorrect material or container shall constitute one defect.
106. Packaging and quantities not as specified.
107. Gross weight exceeds the weight limitation of the box specification for level A or B.
108. Strapping not zinc-coated for level A.
109. Packing not as specified for level A or B.
110. Marking illegible, incorrect, incomplete, or missing.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. The charcoal shall be packaged in quantities of 4, 12½, and 50 pounds as specified, in sacks conforming to PPP-S-50, grade W.

5.2 Packing. Packing shall be level A, B or C as specified.

5.2.1 Level A. The charcoal, packaged as specified in 5.1 shall be packed, with the like weights together, in wood cleated-plywood boxes conforming to PPP-B-601, overseas type, style I or J. Strapping shall be zinc coated and placed adjacent to the skids. Gross weight shall not exceed the weight limitation of the box specification.

5.2.2 Level B. The charcoal, packaged in 4 or 12½ pound quantities as specified in 5.1, shall be packed as specified in 5.2.1 except the boxes shall be domestic type, style optional, and strapping need not be zinc coated. The 50 lb. sacks shall not require overpacking. When specified, the filled sacks shall be palletized in accordance with MIL-STD-147, load type XV.

5.2.3 Level C. The charcoal, packaged as specified in 5.1, shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking.

5.3.1 Military agencies. The charcoal shall be marked in accordance with MIL-STD-129.

5.3.2 Civil agencies. The charcoal shall be marked in accordance with Fed. Std. No. 123.

6. NOTES.

6.1 Intended use. The grade of charcoal specified in this specification is for general foundry use.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

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(a) Title, number, and date of this specification.
(b) Whether lump charcoal or briquet charcoal is required (see 1.2).
(c) Level of packaging and level of packing required (see 5.1 and 5.2).
(d) Quantity to be packaged (see 5.1).
(e) When palletization is required (see 5.2.2).

Custodians:

Army - ME
Air Force - 68

Preparing activity:

Army - ME

Project No. 9110-0005

User activity:

Navy - SA

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Orders for this publication are to be placed with General Services Administration, acting as agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

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